The Man Who Tamed Uncertainty: Andrei Kolmogorov and the Mathematics of Chance*

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February 22, 2026

> 'I, for one, have followed all my life the precept that truth is sacred, that it is our duty to seek it out and to defend it, regardless of whether it is pleasant or not.
>
> Andrei Nikolaevich Kolmogorov
> (25 April 1903 - 20 October 1987)

Figure 1: A.N.Kolmogorov. Source: Wikimedia Commons

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1 Do aliens prefer hotter climates?

Suppose aliens land uniformly at random on a perfectly spherical Earth[1]. By "uniformly," we mean that every patch of equal surface area is equally likely. Now we are told that the landing point lies on a 'great circle.' Is the distribution along that circle uniform?

To answer this, imagine that alien is standing somewhere on Earth, somewhere on the planet's curved surface. Now, suppose you randomly select a point on Earth that lies on the equator. Now, the question takes a simpler form: if you walk along the equator, is every spot equally likely to be the chosen point? The answer seems obvious: yes, of course. Every point on the equator should be equally probable.

Now let us look at the same problem from a different 'angle'. Suppose the chosen point lies on the Prime Meridian, the longitude line running from the North Pole through Greenwich to the South Pole. Now, if instead of the equator you walk along this line, is every spot equally likely? Again, it is an obvious 'yes'.

But thereby hangs a tale: this is the set-up of a famous puzzle that troubled mathematicians in the early 20th century: the equator and the Prime Meridian *intersect*. They share a common point. In fact, any great circle can be described either as a particular latitude or as part of a particular longitude. And yet, when we calculate the probabilities mathematically, we get *different answers* depending on which description we use. This is the Borel-Kolmogorov paradox, named after Émile Borel and Andrei Nikolaevich Kolmogorov. And to understand how Kolmogorov resolved it, and in doing so, transformed probability from a philosophical puzzle into rigorous mathematics, we need to dive into the geometry of spheres and conditional probability.

The Mathematics of the Sphere If you remember your class VII geography, any point on a unit sphere (a sphere with radius 1), can be described any point using two angles. The first is φ (phi), the latitude angle, measuring how far north or south of the equator you are (ranging from $-90°$ to $+90°$). The second is θ (theta), the longitude angle, measuring how far east or west you've traveled (ranging from $0°$ to $360°$). If points are scattered uniformly on the sphere, meaning every patch of equal area is equally likely, the joint probability density of the (lat,lon) turns out to be:

$$f(\varphi, \theta) = \frac{\cos\varphi}{4\pi} \tag{1}$$

The $\cos\varphi$ term on the density comes from simple geometry of circles or spheres. It is easy to see that the circles of latitude aren't all the same size. The equator ($\varphi = 0°$) is the widest circle, with radius 1. But as you move toward the poles, the circles shrink. At $60°$ north latitude, the circle has radius $\cos(60°) = 0.5$, exactly half the size of the equator. At the North Pole ($\varphi = 90°$), the "circle" shrinks to a single point, because $\cos(90°) = 0$. This means a small angular patch near the equator covers *more actual surface area* than the same angular patch near the poles. To keep the distribution uniform over *surface area*, the probability density per degree must be higher near the equator. The $\cos\varphi$ term takes care of that.

Now we can calculate the probability when we 'condition' [2] on being at a particular location. If we condition on being at a fixed longitude (say, $\theta = 0°$), the conditional seems to be:

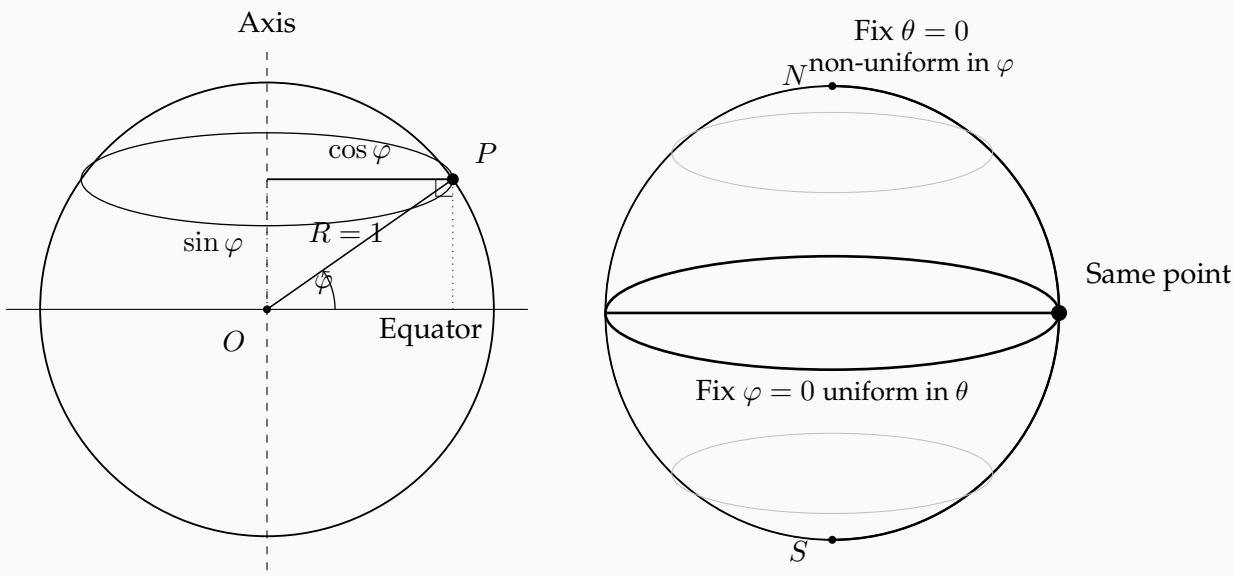
$$f(\varphi \mid \theta = 0°) = \frac{\cos\varphi}{2}, \tag{2}$$

which is *not* uniform! Points are more likely near the equator than near the poles. However, if we condition on being at a fixed latitude, (say, $\varphi = 0°$):

$$f(\theta \mid \varphi = 0°) = \frac{1}{2\pi}, \tag{3}$$

[1] This analogy used in a recent popular account of Kolmogorov's work [Gerovitch, 2023]

[2] In plain English, conditioning means: "How should I update my probabilities given I have some information?"



(a) Latitude circle radius $\cos\varphi$

(b) Same circle, different conditional limits

Figure 2: Geometry of the sphere and the Borel–Kolmogorov paradox.

which *is* uniform: every point around the equator is equally likely. But, you can rotate the sphere and turn the equator into a meridian. That is, a great circle can be the equator can be described as "the circle where $\varphi = 0°$" or equally well as the meridian that is "the intersection of all longitude lines." It's the same circle! How can the probability distribution on it be both uniform and non-uniform? This is the Borel-Kolmogorov paradox [Meehan and Zhang, 2021, Gal, 2014, Krisrs1128, 2013].

If you know a little bit of measure theory, you have probably already identified the issue in my calculation! Simple speaking, when we condition on a zero-probability event (like "being exactly on the equator," which is a one-dimensional curve on a two-dimensional surface and thus has zero area), we're really conditioning on a 'measure-zero' set, and that's an illegal operation!

But, we can still approach the same problem via taking a *limit*, but we need to think carefully about how to approach the limiting event. First, think of horizontal bands getting thinner and thinner around the equator. As the bands shrink toward zero width, we approach '$\varphi = 0°$' and get a uniform distribution. On the other hand, if you take a 'vertical approach', i.e., take vertical slices getting thinner and thinner around a particular longitude. As the slices shrink, we approach "$\theta = 0°$" and get the $\cos\varphi$ distribution.

Different approaches, different limits, different answers, but when you look closer, the conditioning event, 'on the great circle' is *ambiguous* until we specify *how* we're approaching this zero-probability event. One of Kolmogorov's great contributions to probability was to place probability on rigorous mathematical (or analytical) foundations using measure theory, where these subtleties could be precisely defined and understood.

2 The Man Behind the Mathematics

2.1 A Tragic Beginning

Andrei Nikolaevich Kolmogorov was born on April 25, 1903, in Tambov, Russia, about 500 kilometers southeast of Moscow. His birth was marked by tragedy: his unmarried mother, Maria Yakovlevna Kolmogorova, died in childbirth. Little is known about his father, but he was probably named Nikolai Matveyevich Katayev, who was an agronomist involved in

revolutionary politics who disappeared and was likely killed during the Russian Civil War in 1919 [Kendall et al., 1990b, Shiryaev, 1989, Rioul, 2022]. The orphaned child was raised by his two aunts at his grandfather's estate in Tunoshna, near Yaroslavl. His grandfather was a well-to-do nobleman, and young Andrei received an excellent education, first at a village school run by his Aunt Vera.

Even as a child, Kolmogorov showed remarkable mathematical ability. At age five, he was appointed "editor" of the mathematical section of the school journal, charmingly titled *The Swallow of Spring*. At age six, he made his first mathematical discovery: the pattern in sums of odd numbers. It is something any high school student should know: the sum of the first n odd numbers, $1 + 3 + \ldots + (2n - 1)$, always equals n^2. A six-year-old had independently discovered a theorem known to the ancient Greeks. Shiyarev recounts another remarkable story from Kolmogorov's childhood in his interview. To fix a button to a coat, the thread must go through at least two of the button holes, but there could be several ways this could be achieved[3]. As a 5-year old boy, Kolmogorov solved this and arrived at an absolutely correct figure. As a school-student, he invented 'perpetual motion machine' multiple times, and his designs were so elaborate that he was able to fool his school teachers every time.

2.2 Moscow and Mathematical Awakening

In 1910, one of his aunts adopted him and moved to Moscow. He graduated from high school in 1920, during the tumultuous early years of Soviet Russia, and enrolled simultaneously at Moscow State University (studying mathematics and history) and the Mendeleev Institute of Technology (studying metallurgy). At this point, mathematics was not his only love! When he was 10-14 years old, Kolmogorov was deeply fascinated by Biology and Physics, and later in History and Sociology. He apparently thought of and even drafted an utopian constituition for a community that could ensure higher justice in practice. Interestingly, he was captivated by chess, and participated in competitions, but soon, abandoned it forever.

His interest in History was profound, and lifelong, and for a time, the young Kolmogorov was torn between mathematics and history. How he came to choose mathematics remains an interesting, and possible apocryphal story, though repeated across multiple sources.

When he was a 17-year old student, he made his first scientific report to seminar of Prof. S. V. Bakhrushin about the landholding practices in 15th-16th century Novgorod. Prof. Bakhrushin, who was a leading historian, recognized his discovery, and Kolmogorov asked him whether it could be published. When the professor discovered he'd based his analysis on a sample of just five landowners, he said, "You have found only one proof, that is very little for a historian. You need at least five!" Kolmogorov reportedly replied that he'd analyzed all the data available, and one wonders if this early disappointment led him to pursue mathematics where only one proof suffices! [4] Besides drafting a thesis on Russian history, he wrote a treatise on Newtonian mechanics, and somewhere in between also interrupted his studies to become a train conductor, all as a teenager.

Kolmogorov was 17 years old when he entered Moscow State University in an already vibrant and rich atmosphere of young mathematical minds in 1920, and immediately started making breakthroughs on his own, as an undergraduate student. One of Lusin's famous conjecture was that the Fourier series of any square-integrable function would converge almost everywhere. In 1922, at age 19, Kolmogorov provided a brilliant counter-example by constructing a Fourier series that converges to a finite value at every point but diverges "almost everywhere" in a certain technical sense [Kolmogorov, 1923, Wikipedia contributors, 2026b,a].

[3] In his own words, 'There is a button with four holes in it. Thread should go through at least two of them to fix the button. In how many ways can this be done?"

[4] The report he wrote was discovered in his papers, and was published posthumously, and it seems the Kolmogorov did rely on statistical methods for his conclusion. One cannot also help wonder if this brush with the importance of sample size influenced his later work in probability and statistics!

It would take another 40 years to establish the main theorem of harmonic analysis that shows functions of L_p space for $1 < p < \infty$, would have almost everywhere convergence. This remarkable achievement gained him international recognition and probably convinced him to devote his life to mathematics.

He graduated from Moscow State University in 1925, and began his thesis with Nikolai Luzin, and fell under his influence. Luzin was a charismatic mathematician who led a circle of brilliant students they jokingly called "Luzitania", a pun on Luzin's name and the British ocean liner. The students would gather for what they called "joint beating of hearts," passionate discussions of mathematics late into the night. They developed playful terminology: "partial irreverential equations" instead of partial differential equations, "fine night differences" instead of finite differences.

Beginning in 1925, while working on formalizing 'intuitionist logic' with Luzin, Kolmogorov started publishing pathbreaking and foundational papers on probability. This included the famous Kolmogorov inequality that strengthens the famous Tchebychev's inequality, and the Kolmogorov three series theorem that provides the necessary and sufficient conditions for almost sure convergence of a random series $\sum_n X_n$ by requiring convergence of three deterministic series. These results would later inspire the theory of martingales. When Kolmogorov graduated in 1929, he has published 18 papers on logic, analysis, and probability.

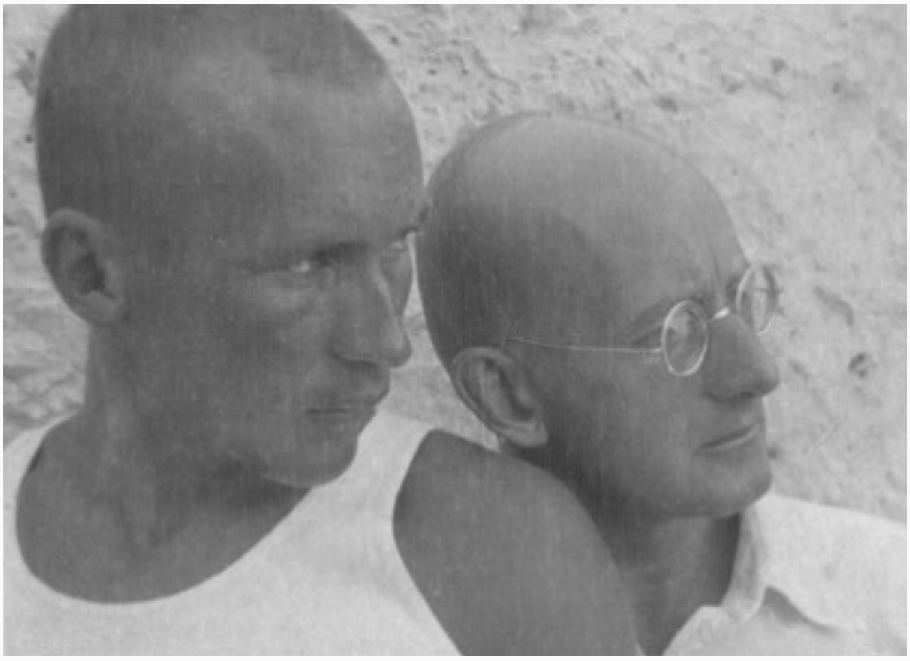


Figure 3: Kolmogorov with Alexandrov. Source: Rioul [2022] as well as Tio Petrus' blog.

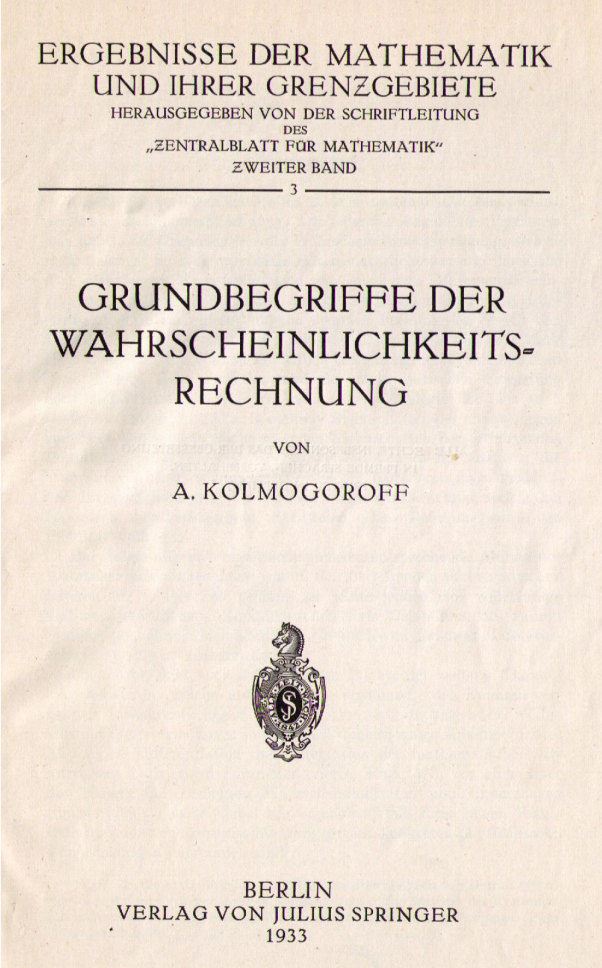
The year 1929 would also be a remarkable epoch in his life: as it marked the beginning of his lifelong friendship with the fellow mathamatician Pavel Alexandrov. That summer they took a long boat trip on the Volga, to lake Sevan in Armenia, and once again in 1931. In 1935, they bought a house together in Komarovka, outside Moscow, where they hosted many renowned mathematicians for discussions. Their relationship was almost certainly romantic, though they lived in a time and place where homosexuality was neither accepted nor openly discussed. This fact would become a vulnerability that Soviet authorities exploited throughout their lives. According to historians Loren Graham and Jean-Michel Kantor, the secret police used knowledge of their relationship to coerce both men into compliance –compelling them to testify against Luzin in 1936, to defend Lysenko's pseudoscience, and later to condemn Solzhenitsyn [Graham and Kantor, 2009, Lorentz, 2002].

3 A Polymath's Legacy

3.1 The Foundations of Probability (1933)

> "It is difficult to overstate the impact of the Grundbegriffe on the development of the subject; essentially the history of probability theory splits in 1933 between pre-Kolmogorov and post-Kolmogorov"
>
> *N. H. Bingham*
> *[Kendall et al., 1990a]*

Kolmogorov's most famous work was his slim 1933 monograph *Grundbegriffe der Wahrscheinlichkeitsrechnung* (Foundations of the Theory of Probability) [Kolmogorov, 1933]. In about 62 pages[5], he transformed probability from a collection of informal methods into a rigorous mathematical theory.


ERGEBNISSE DER MATHEMATIK
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HERAUSGEGEBEN VON DER SCHRIFTLEITUNG
DES
„ZENTRALBLATT FÜR MATHEMATIK"
ZWEITER BAND
3

GRUNDBEGRIFFE DER
WAHRSCHEINLICHKEITS-
RECHNUNG

VON

A. KOLMOGOROFF

BERLIN
VERLAG VON JULIUS SPRINGER
1933

Figure 4: Title page of 'Grundbegriffe der Wahrscheinlichkeitsrechnung' by A.N. Kolmogoroff. Berlin, Springer, 1933. Source: Wikimedia Commons

His approach was simple yet revolutionary: probability is a measure on a set. He defined

[5]The original 1933 Springer monograph had 62 main pages. The 1950 English translation (Foundations of the Theory of Probability, Chelsea) runs longer because of added material and formatting differences, and later reprints may also vary slightly in pagination.

three axioms. First, non-negativity: for any event A, the probability $P(A)$ must be greater than or equal to zero. Second, normalization: the probability of the entire sample space is 1, written as $P(\Omega) = 1$. Third, additivity[6]: for mutually exclusive events A and B, the probability that A or B occurs equals $P(A) + P(B)$.

From these simple axioms, all of probability theory could be derived. It was as if Euclid had returned to organize geometry, but this time, for the mathematics of uncertainty. As Parthasarathy [1988] notes, 'To this day, it has stood the test of time and constitutes the cornerstone around which the entire edifice of statistical theory and computation is erected.'

To appreciate what this achieved, it helps to recall the state of probability before 1933. The classical foundation, dominant for two centuries, defined probability as a ratio of equally likely cases. This worked well for dice and cards. In continuous settings, it became circular or ambiguous. Bertrand's paradoxes (1889) exposed this sharply: ask for the probability that a random chord of a circle exceeds the side length of the inscribed equilateral triangle, and you get different answers depending on what "random" means, with no principled way to choose. The Borel–Kolmogorov paradox in §1 with is a close cousin. Frequentist approaches, championed by von Mises, grounded probability in limiting relative frequencies, but required an idealized infinite sequence as a primitive object. Conditioning on zero-probability events had no rigorous basis in either framework. Borel had introduced countable additivity into probability in 1909, and Fréchet, Steinhaus, Bernstein, and Cantelli had pushed toward measure-theoretic treatments in the decades that followed. But no one had assembled these pieces into a complete, self-contained axiomatic system. As Shafer and Vovk [2006] put it, the *Grundbegriffe* was a work of synthesis as much as of new mathematics: its achievement was to recognize that the right pieces were already on the table, and to take responsibility for declaring the theory complete.

3.2 Random Walks and Markov Processes

In 1931, Kolmogorov published "Analytic Methods in Probability Theory," laying the foundations for the theory of Markov processes [Kolmogorov, 1931]. A Markov process is one where the future depends only on the present state, not on the history of how you got there.

Think of a drunk person taking random steps: at each moment, they step north, south, east, or west with equal probability. Where they step next depends only on where they are now, not on the path they took to get there. This is a random walk, and random walks are everywhere.

Stock prices (approximately) follow random walks. Molecules diffusing through a liquid follow random walks. The electrical signals in your neurons, the fluctuations in population sizes, the spread of diseases, all can be modeled as stochastic (random) processes.

The mathematical framework Kolmogorov developed includes the Chapman-Kolmogorov equations, which describe how probabilities evolve over time:

$$P(i \to k, t_1 + t_2) = \sum_j P(i \to j, t_1) \times P(j \to k, t_2) \tag{4}$$

In words: the probability of getting from state i to state k in time $t_1 + t_2$ equals the sum over all intermediate states j of the probability of going $i \to j$ in time t_1, then $j \to k$ in time t_2. Almost a hundred years later, virtually every domain of applied probability was reshaped by the developments set in motion by the simple, elegant, and fundamental 1931 paper.

[6] One technically important detail: the additivity axiom in the *Grundbegriffe* is σ-additivity (countable additivity), not merely finite additivity. That is, for any countable collection of mutually exclusive events $A_1, A_2, \ldots$, we have $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$. This is the ingredient that makes limit theorems, like the strong law of large numbers, the central limit theorem, derivable within the framework. Finite additivity alone is insufficient.

3.3 Turbulence: The Chaos in Fluids

In 1941, while much of the world was consumed by war, Kolmogorov published revolutionary work on turbulence, the chaotic, swirling motion of fluids.

When you watch smoke rise from a candle, it initially rises smoothly, then breaks into swirling, unpredictable patterns. When you observe a fast-flowing river, you see eddies within eddies, chaos at every scale. This is turbulence, and it's one of the hardest problems in physics.

Kolmogorov developed a statistical theory of turbulence, including what's now called the "Kolmogorov 5/3 law." For turbulent flow at high Reynolds numbers, the energy spectrum follows a power law:

$$E(k) \propto k^{-5/3} \tag{5}$$

where k is the wave number (spatial frequency) and $E(k)$ is the energy at that scale.

This means energy cascades from large eddies to smaller and smaller eddies in a specific mathematical way. The 5/3 law has been verified experimentally in everything from atmospheric flows to ocean currents to laboratory experiments. Once again, nearly 85 years later, Kolmogorov's turbulence theory remains fundamental to fluid dynamics.

3.4 Information and Complexity

During his 1962 visit to the ISI in Kolkata, Kolmogorov developed what we now call *Kolmogorov complexity*, independently discovered around the same time by Ray Solomonoff and Gregory Chaitin. It is another one of his simple-yet-beautiful ideas: the complexity of an object is the length of the shortest computer program that can generate it.

Consider these two sequences of 64 digits:

Sequence A: 0101010101010101010101010101010101010101010101010101010101010101

Sequence B: 1100100101110011100111101011100111101111000010010101101101011011

Which is more complex?

Sequence A can be generated by a simple program: "Print '01' thirty-two times." Very short! Sequence B has no obvious pattern. The shortest program that generates it might be: "Print '1100100101110011...[all 64 digits]...'" The program is as long as the sequence itself. In Kolmogorov's framework, $K(\text{Sequence A})$ is small, it's a simple, compressible pattern. $K(\text{Sequence B})$ is large, it's complex, incompressible, random.

This idea has profound implications. It gives a rigorous definition of randomness. It explains why some things compress well (they're simple) and others don't (they're complex). It's used in data compression algorithms. It provides a measure for how much "information" an object contains. It appears in computational theory, network routing, and cryptography.

Kolmogorov complexity even offers an answer to philosophical questions: What is the difference between a law of nature and a coincidence? A law is a simple pattern (low Kolmogorov complexity) that describes many observations. A coincidence is a complex pattern (high Kolmogorov complexity) with no shorter description.

4 Kolmogorov at ISI, Kolkata

In 1962, Kolmogorov visited the Indian Statistical Institute in Kolkata (then Calcutta) for a month-long stay. The ISI was a remarkable institution, founded in 1931 by P.C. Mahalanobis, it had become a world center for statistics and probability. The story of how Kolmogorov came to India is itself remarkable. In February 1962, the ISI was planning its first convocation. C.R. Rao asked the young probabilists at the institute, S.R.S. Varadhan, K.R. Parthasarathy, and R. Ranga Rao, who they would like to invite from the USSR. The answer was immediate: Kolmogorov.

Rao approached Mahalanobis, who "put the right word in at the right time" to the USSR Academy of Sciences. Kolmogorov agreed to come, but with one condition. He would not fly.

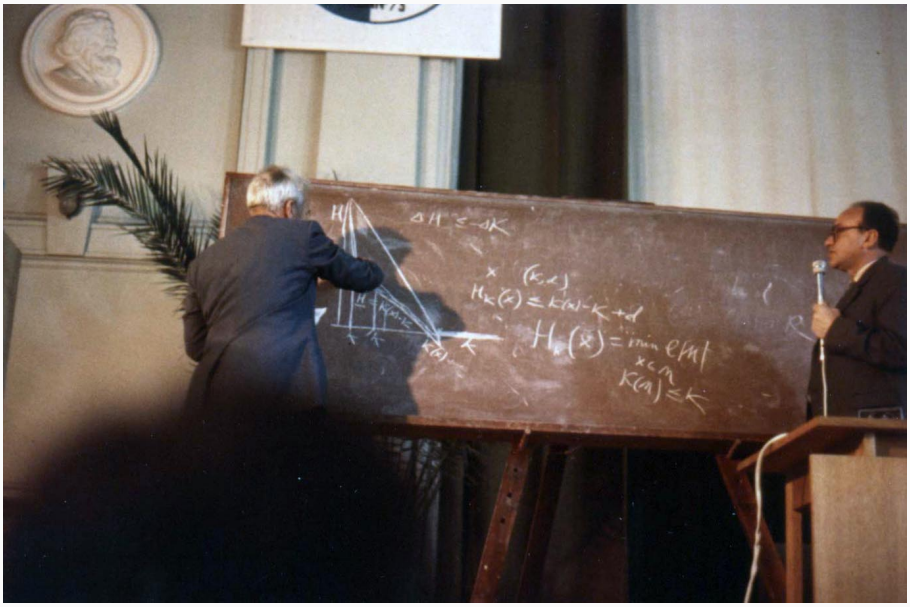

Figure 5: Kolmogorov giving a lecture on Kolmogorov complexity. Source: Wikimedia Commons

Figure 6: P C Mahalanobis welcomes A N Kolmogorov at Amrapali on April 24, 1962. Source: https://www.isid.ac.in/~pcm125/ImageGallery.html

He had problems with his ears and refused to travel by air. But, how does one arrange for a Soviet mathematician to travel by boat from Russia to India in 1962? Kolmogorov found an elegant solution. He would join a Soviet oceanographic expedition, and when the ship reached Bombay (now Mumbai), he would disembark, take a train to Calcutta, and give his lectures at the ISI.

And so he did. During the long sea voyage, Kolmogorov worked on a problem that had fascinated him: ***what is a random number***? How do you define randomness? How do you generate random numbers, and what is the complexity involved in describing them?

On that boat, somewhere on the Indian Ocean, Kolmogorov developed the ideas that would later become *Kolmogorov complexity*, a fundamental concept in computer science and information theory. When he arrived in Calcutta, he prepared a note on his work and submitted it to *Sankhyã*, the ISI's journal, where it was published in 1963 [Parthasarathy, 2019, Kolmogorov, 1963]. Kolmogorov was also an avid swimmer, and the long see voyage allowed him to take a plunge every now and then, for relaxation and stimulation [Parthasarathy, 1988]. Parthasarathy also

recounts that Kolmogorov went on to suggest that institutes and universities in India should be along the coastline, so that the students and faculty can go for a swim before pluning into deeper mathematics.

The institute held its first convocation on February 12, 1962, in a ceremony conducted in the famous mango grove on campus. Five honorary degrees were awarded: to physicist S.N. Bose, to statistician R.A. Fisher, to Prime Minister Jawaharlal Nehru, to quality control pioneer W.A. Shewhart, and to Kolmogorov. A special ceremony for Kolmogorov was held on April 28, 1962, presided over by S.N. Bose himself (Vice-President of the ISI).

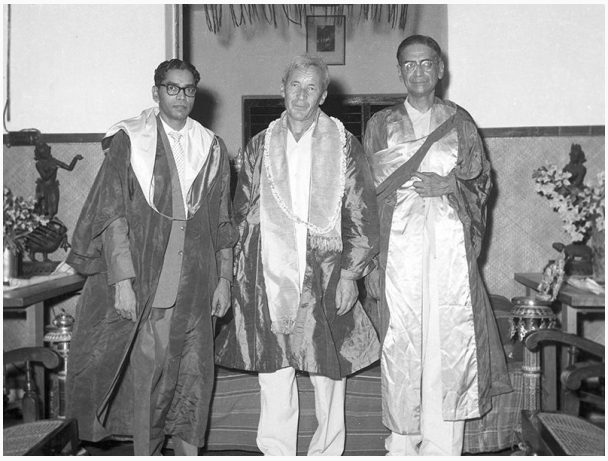

Figure 7: A N Kolmogorov with C R Rao and P C Mahalanobis at the Special Convocation of the Institute on April 28, 1962. Source: https://www.isid.ac.in/~pcm125/ImageGallery.html

The ISI in the 1960s was an extraordinary place. It hosted distinguished visitors from around the world: R.A. Fisher, Norbert Wiener, Yuri Linnik, Jerzy Neyman, Abraham Wald. It had installed the first indigenous computer in India (1953) and operated two of the first digital computers in South Asia, an HEC-2M from England (1956) and a URAL from the Soviet Union (1959). The institute also employed translators who converted Russian mathematical papers into English, reflecting India's position in the Soviet orbit during the Cold War.

There is a funny, possibly anecdotal, story about Kolmogorov's visit, that I first heard from my advisor (Prof. J. K. Ghosh) and later read it on Prof. Debraj Ray's blog [Ray, 2013]. So the story goes that ISI Kolkata had translators among their employees whose job was to translate Russian works in English for the faculty, and one of them were present during the speech. After Kolmogorov spoke, there was awed silence during the question-and-answer session. Finally, the translator stood up and delivered an elaborate homily, in Russian, that went on for a good minute or two:

"*Professor Andrei Nikolaevich Kolmogorov, it is because of the presence of individuals such as yourself that I owe my livelihood. Not just mine, but that of my wife and children, who, but for the grace of your genius, would never have had the opportunities they enjoy...*"

He went on in this vein, a heartfelt tribute to the great mathematician who had made his career as a translator of Russian mathematics possible. When the translator finished, Kolmogorov said: "Excuse me, but my English isn't very good. Could you repeat please?" The translator had been speaking in Russian. Kolmogorov, perhaps not catching that this was meant as a public tribute rather than a question, asked for an English translation of the Russian homage [Ray, 2013].

As K.R. Parthasarathy, one of the probabilists who suggested inviting Kolmogorov, later reflected: "It is interesting in the history of Indian mathematics that at the birth of a very important concept, *Sankhyā* played an important role." Indeed, *Sankhyā* had become a venue where major mathematical ideas first saw print, Arthur Wightman's foundational paper on

Figure 8: Andrei N. Kolmogorov during his 1962 visit to the Indian Statistical Institute, Kolkata. Seated in front; standing behind him (left to right) are K. R. Parthasarathy, B. P. Adhikari, S. R. S. Varadhan, J. Sethuraman, C. R. Rao, and P. K. Pathak. Source: Autobiography of S. R. Srinivasa Varadhan

the axiomatics of quantum field theory also appeared in an Indian journal, the *Journal of Indian Mathematical Society*.

Figure 9: A. N. Kolmogorov Bhavan at the Indian Statistical Institute, Kolkata. Picture courtesy: Prof. Mrinal Kanti Das.

The Abel laureate probabilist S.R.S. Varadhan recalls in a 2018 interview [Zeitouni, 2018] that when Kolmogorov visited the Indian Statistical Institute in 1962, he agreed to serve as an external examiner for his doctoral thesis. At the arranged lecture, which ran far longer than planned, members of the audience began quietly leaving; Kolmogorov, angered by what he perceived as disrespect, threw down the chalk and walked out. He later remarked that in Moscow

seminars ran for hours and that “when Kolmogorov speaks, people should listen" [Zeitouni, 2018]. It is probably worth noting that, at the time of Kolmogorov’s visit, S. R. S. Varadhan, K. R. Parthasarathy, R. Ranga Rao, and J. Sethuraman were young research scholars at ISI, in the early stages of their doctoral work, forming a remarkable cohort that would go on to shape modern probability theory.

To this day, one of the main buildings at ISI Kolkata is named “A.N. Kolmogorov Bhavan” in his honor. If you look closely, there is an “O" missing from the plaque bearing his name.

4.1 A Catalogue of Contributions

Kolmogorov’s influence extended across an astonishing range of fields. He was, undoubtedly, one of the last universalists. When asked if there can be someone like him in the future, versatile in so many fields with so much impact, Shiryaev [1989] lists only four enclyclopedic mathematicians: Poincaré, Hilbert, von Neumann, and Kolmogorov. It is probably worth mentioning this one snippet from [Parthasarathy, 2019]: When the famous American statistician J. Wolfowitz stood up to address the audience at the 1963 conference at Tbilisi, he said: ‘I came to the USSR with the specific purpose of finding out whether Nikolaevich Kolmogorov is an individual or an instituition.’

In pure mathematics, he worked on topology with Pavel Alexandrov, wrote his influential 1925 paper “On the principle of the excluded middle” in intuitionistic logic, and contributed to functional analysis and approximation theory. He made important advances in trigonometric series and developed the Kolmogorov-Arnold-Moser (KAM) theorem in dynamical systems. Working with his student Vladimir Arnold, he produced a partial solution to Hilbert’s 13th problem in 1957 [Arnold, 1993, Tikhomirov, 1988].

In applied mathematics, his work touched celestial mechanics, differential equations, and mathematical linguistics. He developed applications to biology, geology, and metallurgy.

In statistics and probability, he published over 300 research papers and supervised more than 60 PhD students. It’s hard to think of another mathematician of the 20th century with such breadth and depth.

4.2 The Kolmogorov School

Beyond his research, Kolmogorov was passionate about education. In 1963, he founded a specialized mathematical boarding school, Kolmogorov School No. 18 (now the Kolmogorov School at Moscow State University).

He personally taught up to 26 hours per week, leading not just classroom lessons but famous Sunday hikes, 40-kilometer walks with students, filled with mathematical discussions, philosophical debates, and literary conversations. These would end with dinner at his dacha in Komarovka, which he shared with his lifelong friend, the mathematician Pavel Alexandrov.

Kolmogorov reformed the mathematics curriculum across the Soviet Union and wrote textbooks for grades 6–10 that influenced mathematical education for generations. He contributed over 80 articles to the Great Soviet Encyclopedia, making advanced mathematics accessible to general readers.

5 The Soviet Context: Mathematics Under Stalin

5.1 The Luzin Affair

In 1936, Kolmogorov faced one of the darkest episodes of his life. His beloved teacher Nikolai Luzin was accused of plagiarism and of being a “servant to fascistoid science” and an enemy of the Soviet people. This was during Stalin’s Great Purge, when such accusations typically led to imprisonment or execution. [Lorentz, 2002, Rioul, 2022, Khimchenko, 2001]

Figure 10: Nikolai Luzin (9 December 1883 – 28 February 1950). Source: Wikimedia Commons

Kolmogorov and other former students were called to testify. They did testify against Luzin. Whether they were coerced, whether they believed the charges, or whether personal acrimony played a role remains a matter of historical debate. None of the students ever spoke publicly about the affair afterward [Arnold, 1993, Shiryaev, 1989]. Curiously, Luzin was not arrested or expelled from the Academy of Sciences. He lost some positions but continued working. Recent archival evidence from the 1990s suggests that Stalin personally concluded Luzin posed no real threat.

The affair left its mark on Kolmogorov. He became known as one of the very few "non-political mathematicians with real power" in the Soviet system, someone who could navigate between scientific discovery and political constraint, advancing mathematics while avoiding the fate of so many intellectuals in Stalin's Russia.

The bitterness lingered for years; in 1946, after Luzin voted against Alexandrov's election to the Academy of Sciences, Kolmogorov publicly slapped his former teacher—an episode later reported to the Kremlin and, remarkably, treated with relative indulgence by Stalin [Rioul, 2022, Lorentz, 2002].

It would be too simple, however, to portray Kolmogorov only as a victim of ideological coercion. His relationship with Luzin was already strained well before the events of 1936. In his last interview [Khimchenko, 2001], Kolmogorov recalled how Luzin had, with great insistence, convinced the young Pavel Alexandrov that he was destined to solve the Continuum Problem, planting in him an expectation so heavy that it led to a personal crisis and a temporary desire to abandon mathematics.

Personal tensions and intellectual disagreements thus predated the political drama. The subsequent rupture cannot be explained solely by external pressure; private loyalties, wounded pride, and generational conflict were already entangled with the coming storm.

5.2 Science in Service of the State

The Soviet state demanded that science serve practical purposes. The official stance was that the government had all correct answers; competing ideas were rejected. Yet somehow, Kolmogorov and his students developed probability and statistics into powerful tools.

At the height of Lysenkoism, Soviet biology was reshaped by the doctrine that heredity could be molded by environment and that classical Mendelian genetics was a "bourgeois" fraud.

Lysenko's slogan that "**science is the enemy of chance**" expressed not merely a biological claim but an ideological suspicion toward probabilistic reasoning itself [Rioul, 2022]. In February 1940 Kolmogorov published a short note defending Mendelian laws, arguing on mathematical grounds that the experimental claims advanced by Lysenko's supporters were fully consistent with classical genetics; when Lysenkoism became official doctrine in 1948, he was compelled to retract publicly [Rioul, 2022]. Ideological scrutiny extended beyond biology. After a 1936 *Pravda* denunciation of Soviet mathematicians for publishing their best work abroad, Kolmogorov arranged for a Russian edition of the *Grundbegriffe*, and in the ensuing years major scientific work increasingly appeared in Soviet journals [Rioul, 2022].

Yet, as G. G. Lorentz later remarked, the position of mathematicians, precarious though it could be, was often less catastrophic than that of writers and poets, whose work was more immediately exposed to political interpretation [Lorentz, 2002]. The broader Stalinist period had already witnessed the execution of Nikolai Gumilev in 1921, the death of Osip Mandelstam in a transit camp in 1938, and the suicide of Marina Tsvetaeva in 1941; Vladimir Mayakovsky had taken his own life in 1930. Mathematics, by contrast, sometimes afforded a narrow shelter in abstraction, even if it could not entirely shield its practitioners from ideological pressure. Even within mathematics, ideological pressure did not entirely spare personal relationships: during the 1936 Luzin affair, Kolmogorov testified to Luzin's scientific achievements while simultaneously referring to his alleged "moral and political decadence," a formulation that illustrates the uneasy balance between loyalty and conformity [Rioul, 2022].

During World War II, Kolmogorov applied his statistical theories to artillery fire and developed a stochastic distribution scheme for barrage balloons to protect Moscow during the fierce Battle of Moscow. His 1938 paper on "smoothing and predicting stationary stochastic processes" would later find major applications during the Cold War, though these were, of course, classified.

His contributions were recognized even within the Soviet system. In 1939, he was elected Academician of the USSR Academy of Sciences. In 1941, he received the Stalin Prize, followed by the Lenin Prize in 1965. Later, international recognition followed: the Wolf Prize in 1980, one of the highest honors in mathematics, and the Lobachevsky Prize in 1987.

6 Pavel Alexandrov: Partnership and Protection

> "In the country house where P.S. Alexandrov and A.N. Kolmogorov held their famous gatherings for many years, there was a seminar room with a blackboard. Several years after Kolmogorov's death it had not been erased, and still had in his hand, in English, the motto:
>
> **MEN ARE CRUEL, BUT MAN IS KIND.**"
>
> Khimchenko [2001]

Throughout his life, Kolmogorov's closest relationship was with mathematician Pavel Alexandrov. The friendship that started during a boat trip on the Volga, would last 53 years, until Kolmogorov's death. In 1935, Alexandrov and Kolmogorov bought a house together in Komarovka, outside Moscow. They invited many renowned mathematicians there for discussions. A year before his death, Kolmogorov confided: "For me, these 53 years of intimate and indissoluble friendship were the reason why my whole life was filled with happiness, and

the basis of this happiness was the permanent consideration that Alexandrov made for me" [Rioul, 2022].

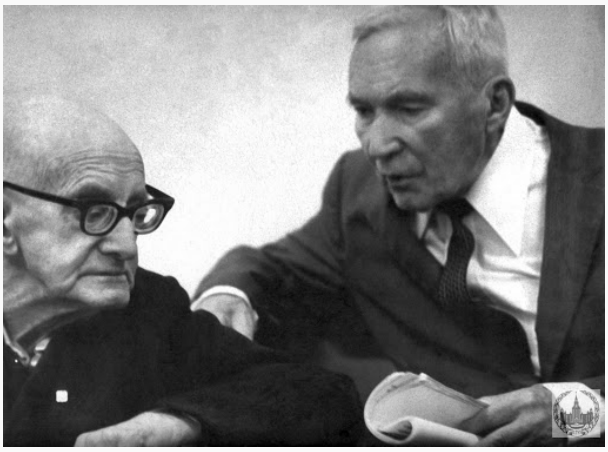

Figure 11: Alexandrov and Kolmogorov. Source: Chronicle of Moscow University

Their relationship was almost certainly romantic, though they lived in a time and place where homosexuality was not accepted, let alone openly discussed. This fact would become a vulnerability that the Soviet secret police exploited repeatedly throughout their lives. According to historians Loren Graham and Jean-Michel Kantor, the Soviet secret police gathered information on all prominent people, including scholars, noting their sexual and personal habits. The police soon learned of Kolmogorov and Alexandrov's relationship, and they used that knowledge to obtain the behavior they wished [Graham and Kantor, 2009].

When the police asked Kolmogorov and Alexandrov to join in attacking Luzin in 1936, they did so. When the government asked them to defend the pseudo-scientist Trofim Lysenko (even though Kolmogorov had earlier criticized the biologist), they complied. When, after World War II, they were asked to write a condemnation of Alexander Solzhenitsyn in *Pravda*, calling him a traitor, they published such a joint letter.

NO PARDON FOR TREASON

"We learned with deep satisfaction that Solzhenitsyn has been deprived of Soviet citizenship and bounced out of our country. Soviet intelligentsia are characterized above all by their high civic consciousness, feeling of duty to people and state, respect for traditions and emblems of the people and pride in the high achievements of the Soviet people building communism. In his creations, published in the West, A. Solzhenitsyn blackens our social structure, desecrates the memory of those fallen in the battles of the Great Patriotic War and purposefully gives a distorted picture of the life of Soviet people. In his way he not only violates Soviet laws but also trespasses in the inner sanctum of our people. He has put himself outside of our society. Such persons have no place in our country."

Excerpt from a joint letter by Pavel Alexandrov and Andrei Kolmogorov condemning Alexander Solzhenitsyn, published in Pravda, *February 16, 1974. The letter was written under pressure from Soviet authorities, who used blackmail to compel the mathematicians' participation.*

Kolmogorov on several occasions tried to explain his inconsistencies and what must have seemed like disloyalties to colleagues, saying, "Sometime I will explain everything to you." Shortly before his death he stated that he would "fear 'them' [the secret police] to his last day" [Graham and Kantor, 2009]. This might help the reader today to make sense of what otherwise seem inexplicable: how could one of the greatest minds of the century participate in attacking his mentor, supporting junk science, and condemning a courageous dissident?

The answer appears to be that Kolmogorov valued his relationship with Alexandrov above nearly everything else, and the Soviet authorities knew it. They held this relationship hostage, and Kolmogorov paid the price they demanded to protect it, and to protect Alexandrov.

6.1 The Moral Weight of Choices

It is easy, from the safety of our time and place, to judge the choices Kolmogorov made. Theoretical computer scientist Scott Aaronson has written thoughtfully about what he calls "the Kolmogorov option": the strategy of building up fortresses of truth in places the ideological authorities don't particularly understand or care about, like pure mathematics, while avoiding direct assault on beliefs your culture considers necessary for it to operate [Aaronson, 2017].

As Aaronson points out, Kolmogorov knew better than to pick fights he couldn't win. He judged that he could best serve the cause of truth by building up an enclosed bubble of truth and protecting that bubble from interference by the Soviet system, and even making the bubble useful to the system wherever he could, rather than futilely struggling to reform the system and simply making martyrs of himself and all his students.

This was not pure cowardice, nor was it pure collaboration. It was a third path, one of strategic accommodation and selective resistance. When Kolmogorov could safely advance truth, he did so. His 1940 paper courageously supporting Mendelian genetics, even when Lysenkoism was ascendant, showed he was willing to take risks when the stakes were high and there was a chance of making a difference. But he also made compromises that haunt his legacy. The attack on Luzin. The letter condemning Solzhenitsyn. These were not just passive acquiescence, but active participation in injustice.

Perhaps the fairest judgment is the one suggested by Olivier Rioul: that Kolmogorov was "a particularly smiling man, open and generous especially with his students," who "did not hesitate to encourage young talented mathematicians, in Russia or abroad." He went to great lengths to advance them in their careers, not hesitating to finance their stays out of his personal funds. He established schools for mathematically gifted children. He quietly protected Jewish researchers during periods of intense antisemitism in Soviet mathematics [Rioul, 2022].

These were not the actions of a purely self-interested collaborator, but rather of someone trying to do good within impossibly constrained circumstances. The Soviet authorities put him in situations where there was no choice that wouldn't involve betraying someone or something he cared about. He chose to protect Alexandrov, his students, his ability to do mathematics. Others might have made different choices. Some, like Andrei Sakharov, did.

There is no simple moral to this story. Kolmogorov was neither hero nor villain, but a brilliant, flawed human being navigating an inhuman system. His mathematical legacy is secure. His moral legacy is complicated. Perhaps that is exactly as it should be: a reminder that genius does not automatically confer moral courage, and that even the greatest minds face agonizing choices when living under tyranny.

7 The Human Side

Parthasarathy described Kolmogorov as "a great humanist," recalling how he was visibly moved by the poverty he witnessed in India, photographed ordinary people at work, reflected publicly on the coexistence of poverty and plenty, and showed deep concern for children and their education [Parthasarathy, 1988, p. 448]. Parthasarathy also remembered him as "an outdoor mathematician," to whom key ideas came while walking in the woods, swimming in the sea or a lake, rowing, or skiing down mountain slopes [Parthasarathy, 1988, p. 445]. Mathematics for Kolmogorov was not confined to the desk; it unfolded in motion, in landscape, and in conversation.

Kolmogorov never spoke about his personal life in public, but we know he married Anna Dmitrievna Egorova in 1942. His relationship with Pavel Alexandrov, with whom he bought a house in 1935 and lived for decades, was likely more than friendship, though this remains a matter of historical interpretation.

What is clear is that Kolmogorov was a man of broad cultural interests: history, literature, philosophy, and poetry. He was not a narrow specialist but a Renaissance figure who saw mathematics as part of the larger tapestry of human knowledge. In 1971, at age 68, he joined an oceanographic expedition aboard the research vessel *Dmitri Mendeleev*. He worked with students well into old age, even as Parkinson's disease took its toll. In his final years, nearly blind, he continued thinking about mathematics.

He died on October 20, 1987, in Moscow, and was buried in Novodevichy Cemetery, the resting place of many of Russia's most celebrated artists, writers, and scientists.

8 Coda: The Harmony of Uncertainty

Figure 12: Source: Video library, 'On the centenary of the great Russian scientist Andrei Nikolaevich Kolmogorov (25.IV.1903–20.X.1987)'.

Vladimir Arnold, one of Kolmogorov's most famous students, once said: "Kolmogorov, Poincaré, Gauss, Euler, Newton, are only five lives separating us from the source of our science" [Arnold, 1993]. Five lives. From Newton, who created calculus and discovered the laws of motion and gravitation, to Kolmogorov, who created the mathematics of randomness and uncertainty, just five lifetimes span the distance.

There's something appropriate about this. Kolmogorov spent his life showing that randomness isn't the absence of pattern but a deeper kind of order. The Borel-Kolmogorov paradox, far from being a flaw in mathematics, revealed the subtle structure underlying probability. Turbulent fluids, seemingly chaotic, follow precise statistical laws. Random sequences can be defined by their complexity. Uncertainty itself can be axiomatized. Kolmogorov believed, as he once said, that "every mathematician believes that he is ahead of the others. The reason none state this belief in public is because they are intelligent people."

Perhaps he was ahead. Perhaps his genius lay in seeing that the universe, even in its randomness, has an inner harmony, and that mathematics could reveal it.

From a sphere's surface to the flow of fluids, from the random walk of particles to the complexity of information, from the tragedy of his birth to the triumph of his ideas, Kolmogorov's life was itself a kind of proof: that even in the midst of chaos, uncertainty, and historical turbulence, the human mind can find patterns, create beauty, and reveal truth.

Personally, I would like to believe that he really was ahead of his time, and that he foresaw that even the technological future, too, shall pass.

I will close with a small snippet from his last interview with film-maker Aleksandr Nikolaevich Marutyan, for his 1983 film "Stories on Kolmogorov":

> **K.:** Are you familiar with Shklovskii's *The Universe, Life, and the Mind*?
> **M.:** Yes.
> **K.:** He maintains that the development of every culture, if it is not aborted by some catastrophic events—and we all know what might befall humankind now—culminates in a stage of *loss of interest in technology*. Perhaps he really is right.
> **M.:** What does "loss of interest in technology" mean? You mean that people occupy themselves more with humanistic problems?
> **K.:** Not really humanistic problems. But it must be possible to return to a more basic and child-like joy in living. Do you know the German writer Hesse?
> **M.:** Yes.
> **K.:** In *Das Glasperlenspiel*, Hesse depicts such a society, and quite brilliantly, I would say. A society which has lost interest in technological progress.

Appendix

Key Events

1903–1920: Childhood & Revolution

- Born April 25, 1903 in Tambov. Mother dies in childbirth.
- Father disappears in Civil War (1919).
- Age 6: Discovers pattern $1 + 3 + 5 + \cdots + (2n - 1) = n^2$.
- Raised by aunts at grandfather's estate.

1920–1935: Student & Rising Star

- **1920**: Enters Moscow State University.
- **1922** (age 19): Fourier series diverging almost everywhere $\rightarrow$ international fame.
- **1925**: 8 papers including intuitionist logic, probability theory.
- **1929**: Meets Pavel Alexandrov – 58-year partnership begins.
- **1931**: Professor. Publishes Markov processes, Chapman-Kolmogorov equation.
- **1933**: *Grundbegriffe der Wahrscheinlichkeitsrechnung* – revolutionizes probability with three axioms.

1936–1945: Terror & War

- **1936**: **Luzin Affair** – blackmailed via relationship with Alexandrov, testifies against mentor. Stalin's Great Purge begins.
- **1940**: Courageously defends Mendelian genetics against Lysenkoism.
- **1941**: Germany invades USSR. Publishes turbulence theory (K41, 5/3 law). Applies math to Moscow defense.
- **1941–44**: Siege of Leningrad (872 days, $\sim$1 million dead).
- **1942**: Marries Anna Dmitrievna Egorova.

1945–1962: Post-War Achievements

- **1948**: Lysenko triumphs – forced to retract genetics support.
- **1950s**: Introduces ε-entropy. KAM theorem (dynamical systems).
- **1953**: Stalin dies – gradual thaw.
- **1957**: With Arnold, solves Hilbert's 13th problem.
- **1962**: Visits ISI Kolkata by ship (develops Kolmogorov complexity during voyage). Publishes in *Sankhyā* (1963).

1963–1987: Elder Statesman

- **1963**: Founds School No. 18 for mathematically gifted children.
- **1970s**: Sophistication theory (refinement of Kolmogorov complexity).
- **1974**: With Alexandrov, signs *Pravda* letter condemning Solzhenitsyn (under pressure).
- **1980**: Wolf Prize.
- **1987**: Dies October 20 (age 84) from Parkinson's disease.
- **1991**: Soviet Union dissolves (4 years after his death).

Major Historical Events He Lived Through

WWI (1914–18)	Age 11–15
Russian Revolution (1917)	Age 14
Civil War (1918–21)	Age 15–18; father killed
Stalin's Purges (1936–38)	Age 33–35; directly affected
WWII (1941–45)	Age 38–42; applied work for defense
Siege of Leningrad (1941–44)	Age 38–41
Stalin's Death (1953)	Age 50
Cold War (1947–91)	Age 44–84; entire mature career

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